Exercise 4

For a data set $\mathcal{D} = ((X_1, Y_1), \ldots, (X_m, Y_m)) \in (\mathcal{X} \times \mathcal{Y})^m$, where $\mathcal{Y} = \mathbb{R}$, and any permutation $\sigma \in S_m$, put

$$\mathcal{D}_\sigma = ((X_{\sigma(1)}, Y_{\sigma(1)}), \ldots, (X_{\sigma(m)}, Y_{\sigma(m)})).$$

Given model $\hat{f}(x; \mathcal{D})$, we construct a randomized score function $s((x, y); \mathcal{D}, \xi)$, where ξ is a random variable independent of the data set, as follows. Let σ_ξ be an S_m-valued random variable depending only on ξ that has the uniform distribution on S_m if ξ has the uniform distribution on $[0, 1]$. Put

$$s((x, y); \mathcal{D}, \xi) = \left| y - \hat{f}(x; \mathcal{D}_{\sigma_\xi}) \right|$$

for all (x, y) and all values of ξ. Prove that s is a symmetric score function in the following sense: for every fixed test points $(x_1^{'}, y_1^{'}), \cdots, (x_{m'}^{'}, y_{m'}^{'})$ and every $\tau \in S_m$, the vector

$$\left(s\left((x_1^{'}, y_1^{'}); \mathcal{D}, \xi\right), \ldots, s\left((x_{m'}^{'}, y_{m'}^{'}); \mathcal{D}, \xi\right) \right) \in \mathbb{R}^{m'}$$

has the same distribution as

$$\left(s\left((x_1^{'}, y_1^{'}); \mathcal{D}_\tau, \xi\right), \ldots, s\left((x_{m'}^{'}, y_{m'}^{'}); \mathcal{D}_\tau, \xi\right) \right) \in \mathbb{R}^{m'}$$

when ξ has the uniform distribution on $[0, 1]$.

Solution For each $i \in \{1, \ldots, m'\}$, the random variable

$$s\left((x_i^{'}, y_i^{'}); \mathcal{D}, \xi\right) = \left| y_i^{'} - \hat{f}(x_i^{'}; \mathcal{D}_{\sigma_\xi}) \right|$$

has the same distribution as

$$\left| y_i^{'} - \hat{f}(x_i^{'}; \mathcal{D}_{\tau \circ \sigma_\xi}) \right| = s\left((x_i^{'}, y_i^{'}); \mathcal{D}_\tau, \xi\right)$$

as long as ξ has the uniform distribution on $[0, 1]$ because, by construction, both σ_ξ and $\tau \circ \sigma_\xi$ have uniform distribution on S_m. ////